

SWINWAVED: MULTI-RESOLUTION WHALE-CALL DETECTION WITH CALIBRATION AND WAVELET-RIDGE VERIFICATION

Technical Report

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ABSTRACT

This report describes a system for BioDCASE 2026 Task 2, supervised detection of strongly-labelled Antarctic blue and fin whale calls. Our SwinWaveD acoustic detector adapts the 2025 winning multi-resolution learnable time-frequency/Swin Transformer architecture to the 2026 data. We replace the external LEAF dependency by a self-contained learnable complex Gabor/PCEN frontend, use three parallel Swin branches, and perform dense frame-level multi-label detection. The main engineering contribution is systematic event-level calibration: fine-label thresholds, smoothing, temporal merging, duration constraints, and non-maximum suppression. A second contribution is a post-hoc Morlet wavelet-ridge verifier applied only to D-call candidates to suppress false positives. On the official 2026 validation split, fine-label calibration achieved macro-F1 0.3533 and micro-F1 0.3840. Adding the D-call wavelet verifier improved macro-F1 to 0.3583 and micro-F1 to 0.3974 by reducing D-call false positives from 8514 to 6673. Source code is available at https://github.com/hugomagaldi/BioDCASE2026_WhaleCalls.

Index Terms— Bioacoustics, whale-call detection, Swin Transformer, PCEN, wavelet verification

1. INTRODUCTION

BioDCASE 2026 Task 2 is a strongly labelled, multi-class and multi-label sound event detection task for seven Antarctic blue and fin whale call types. Systems must estimate both call labels and temporal boundaries. The seven labels are evaluated in three groups: BmABZ, D, and Bp [1]. The development set contains 6591 recordings sampled at 250 Hz; Casey2017, Kerguelen2014, and Kerguelen2015 form the official validation split [1]. Because calls are sparse and soundscapes vary strongly across site-years, the task stresses generalisation and false-positive control.

Our starting point is the 2025 winning method of Marolt and Bones, which used three multi-resolution LEAF representations, three parallel Swin Transformers, feature-map fusion, and shifted-window averaging [2, 3]. The present system keeps this architectural family but adapts it to the 2026 task with a self-contained learnable frontend, extensive calibration, and a lightweight wavelet-ridge verifier that targets the D-call false-positive bottleneck.

2. NEURAL DETECTOR

Input and frontend. Audio is read at 250 Hz and split into 16 s windows. Each crop is normalised independently. The frontend

contains three learnable complex Gabor filterbanks with 64 filters each, frequency range 5–124 Hz, and kernel lengths 65, 129, and 257 samples. The complex magnitude is pooled to a 0.1 s frame hop and compressed with a PCEN-like channel-wise normalisation. This approximates the LEAF idea while avoiding a dependency on an external LEAF implementation.

Backbone. Each resolution is processed by one Swin branch [4]. The implementation uses embedding dimension 32, branch depths (1, 1, 1), attention heads (2, 4, 4), local window size 4, and hierarchical patch merging. Branch feature maps are resized to a common resolution, concatenated, and passed through a 128-channel convolutional fusion block. The fused representation is up-sampled to the 0.1 s grid, averaged over frequency, and decoded into seven sigmoid logits.

Training. Random 16 s crops are drawn from the eight training site-years. Positive-event oversampling (ratio 0.7) mitigates sparsity. Annotations are converted to frame-level binary targets for the seven fine labels, and the loss is weighted binary cross-entropy with positive weights estimated from annotated duration. Training uses AdamW, learning rate 2×10^{-4} , weight decay 10^{-4} , batch size 16, 40 epochs, 3000 steps/epoch, gradient clipping at 5, mixed precision, and seed 1337. The acoustic model has 1,108,935 trainable parameters, about 4.23 MB in fp32.

3. EVENT DECODING AND CALIBRATION

At inference, 16 s windows are evaluated with 0.5 s stride. Frame probabilities from all windows covering the same time are averaged. The event decoder applies a 5-frame moving average, fine-label thresholds, gap merging up to 0.5 s, duration limits from 0.25 s to 30 s, and within-class temporal non-maximum suppression. This stage is important because the official event metric penalises duplicate detections overlapping a single ground-truth event [1].

A single global threshold gave high recall but excessive false positives. We therefore tuned thresholds independently for the seven fine labels on the official validation site-years, using the official evaluator and macro-F1 as the objective. The selected thresholds were BmA 0.930, BmB 0.990, BmZ 0.995, BmD 0.995, BpD 0.930, Bp20 0.960, and Bp20plus 0.985.

4. WAVELET-RIDGE VERIFIER

After calibration, the D group remained the main low-precision class. We added a post-hoc verifier for BmD and BpD candidates only, leaving BmABZ and Bp predictions unchanged. For each D candidate, a complex Morlet continuous wavelet transform [5] is

Table 1: Validation results on Casey2017, Kerguelen2014, and Kerguelen2015. “Cal.” denotes fine-label calibration; “Wav.” adds D-call wavelet verification.

Sys.	Group	TP	FP	FN	Prec.	Rec.	F1
Cal.	BmABZ	5711	7951	3752	0.4180	0.6035	0.4939
Cal.	D	972	8514	1884	0.1025	0.3403	0.1575
Cal.	Bp	2122	2961	3185	0.4175	0.3998	0.4085
Cal.	Micro	8805	19426	8821	0.3119	0.4995	0.3840
Wav.	BmABZ	5711	7951	3752	0.4180	0.6035	0.4939
Wav.	D	899	6673	1957	0.1187	0.3148	0.1724
Wav.	Bp	2122	2961	3185	0.4175	0.3998	0.4085
Wav.	Micro	8732	17585	8894	0.3318	0.4954	0.3974

computed with 2 s context and logarithmically spaced frequencies over 12–125 Hz. Evidence is measured in the expected 20–120 Hz D-call band, consistent with the task call descriptions [1].

Let $P(f, t)$ be wavelet power, e the candidate event interval, c the local context outside the event, and B the D-call band. We compute a band-energy ratio r_b , ridge concentration r_r , and contrast score r_c :

$$r_b = \frac{\sum_{f \in B, t \in e} P(f, t)}{\sum_{f, t \in e} P(f, t)}, \quad r_r = \text{mean}_{t \in e} \frac{\max_{f \in B} P(f, t)}{\sum_{f \in B} P(f, t)}, \quad (1)$$

$$r_c = \sigma \left(\frac{10 \log_{10}(\bar{P}_{B,e} / \bar{P}_{B,c})}{6} \right), \quad s(e) = 0.45r_b + 0.35r_r + 0.20r_c. \quad (2)$$

Validation grid search selected $s(e) \geq 0.3797$ for accepting D-call candidates. The verifier is interpretable and requires no extra training.

5. VALIDATION RESULTS

Table 1 reports validation performance. Fine-label calibration produced a macro-F1 of 0.3533 and micro-F1 of 0.3840. The wavelet verifier reduced D false positives from 8514 to 6673 while retaining 899 true positives, increasing D F1 from 0.1575 to 0.1724. Since the verifier is D-only, BmABZ and Bp are unchanged. The final system reaches macro-F1 0.3583 and micro-F1 0.3974.

6. SUBMISSION AND REPRODUCIBILITY

No external datasets or pretrained models were used. The final system is the calibrated neural detector followed by D-call wavelet verification at threshold 0.3797. The public implementation, including inference scripts and meta-data, is available at https://github.com/hugomagaldi/BioDCASE2026_WhaleCalls.

7. CONCLUSION

The submitted method is a 2026 adaptation of the strongest 2025 architectural family, but its main performance gains come from event-level calibration and targeted false-positive suppression. Fine-label thresholding establishes a strong operating point for BmABZ and Bp. The D-call wavelet-ridge verifier provides a measurable

improvement without retraining, adding an interpretable signal-processing prior to a learned detector. Remaining errors are dominated by D-call confusions, especially false positives in difficult site-years.

8. REFERENCES

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